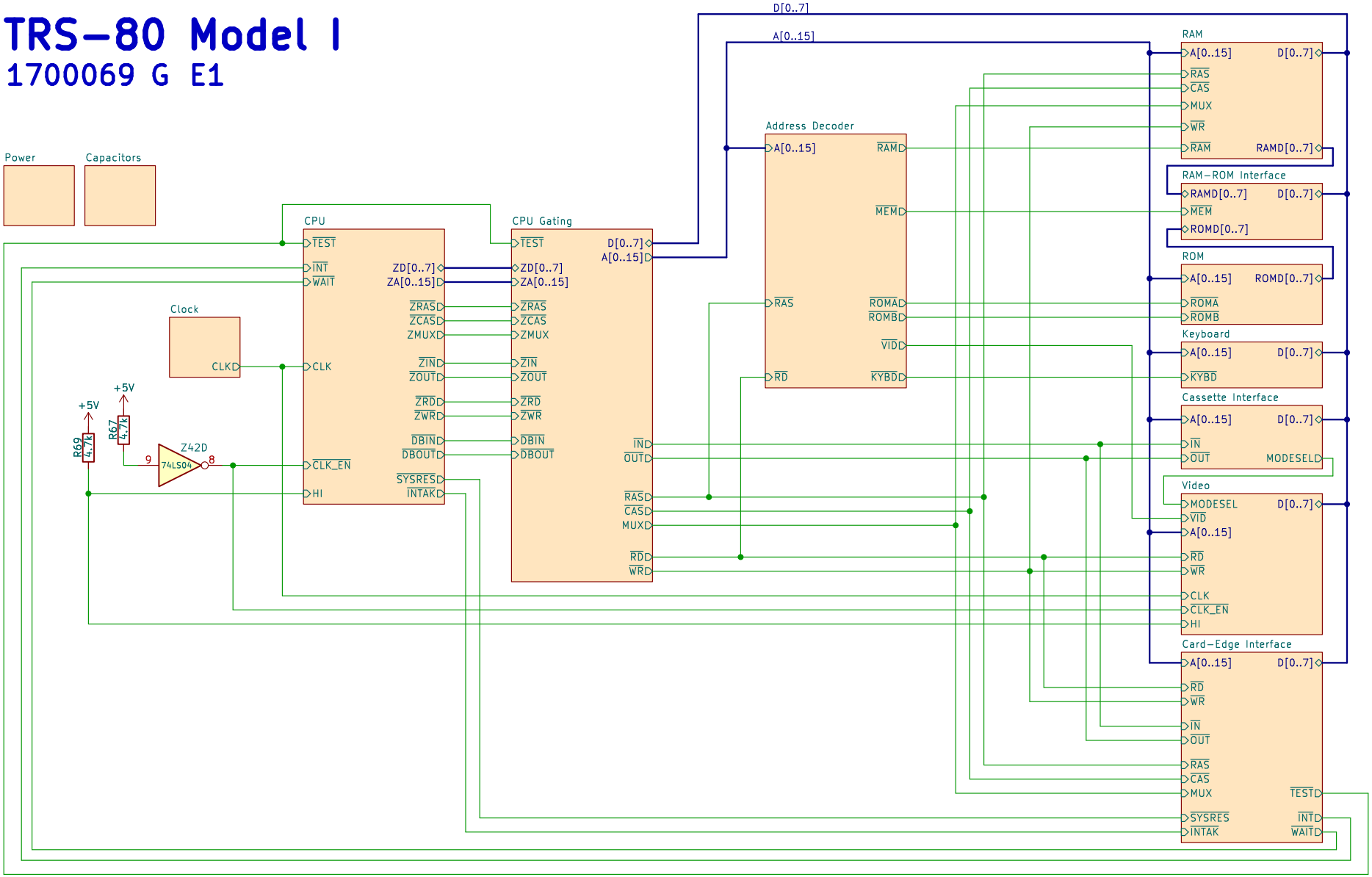
TRS-80 Model I
1700069 G E1



System Overview

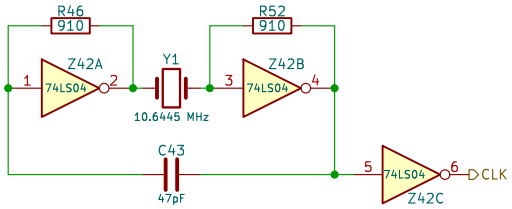
Full overview of the system

RetroStack – Marcel Erz

Sheet: /
File: TRS80_Model_I_G_E1.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 1/21



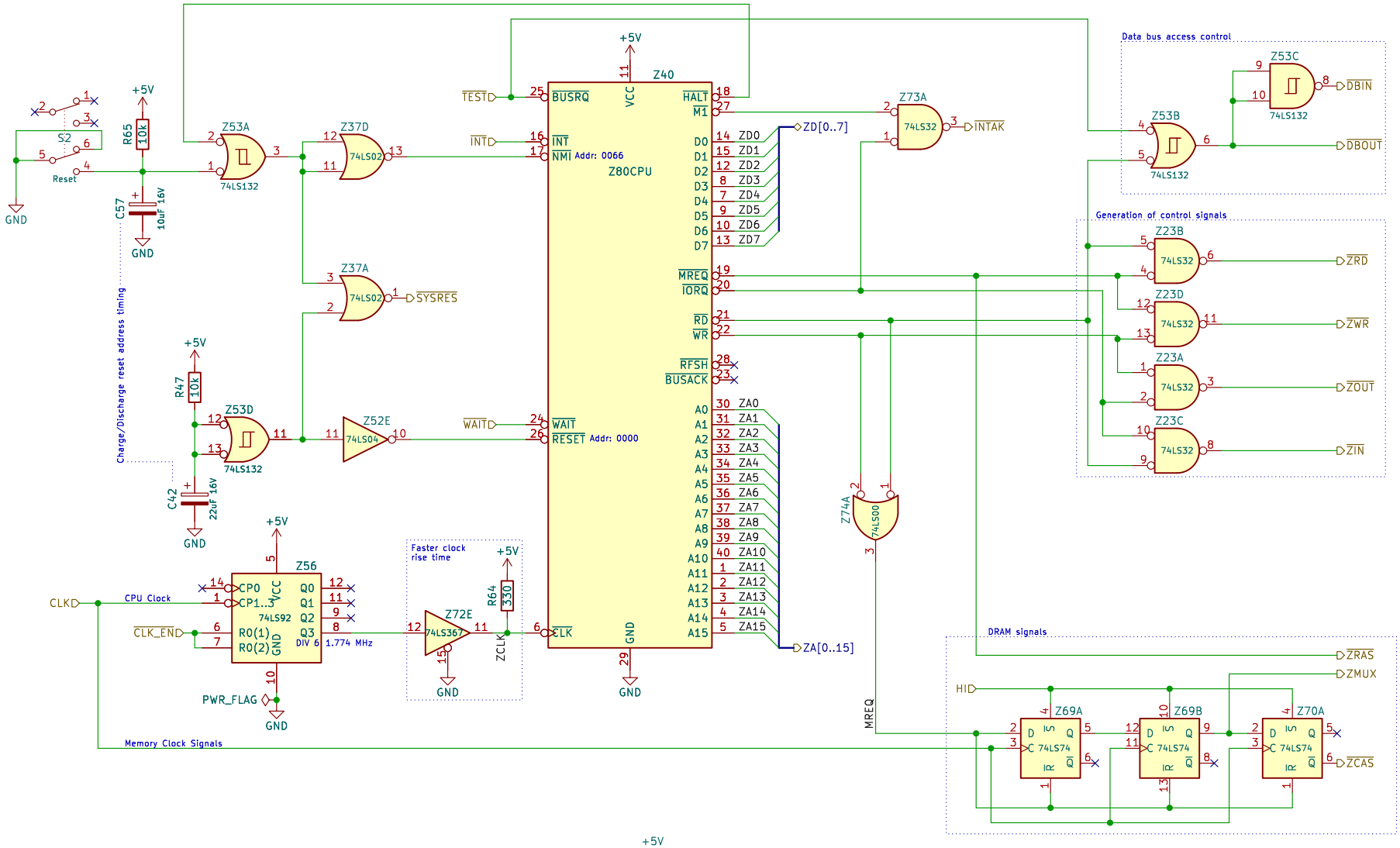
System Clock

RetroStack – Marcel Erz

Sheet: /Clock/
File: Clock.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 2/21



CPU & Signals

CPU with reset circuits (button and power-up), clock divider, RAS & CAS divider, signals

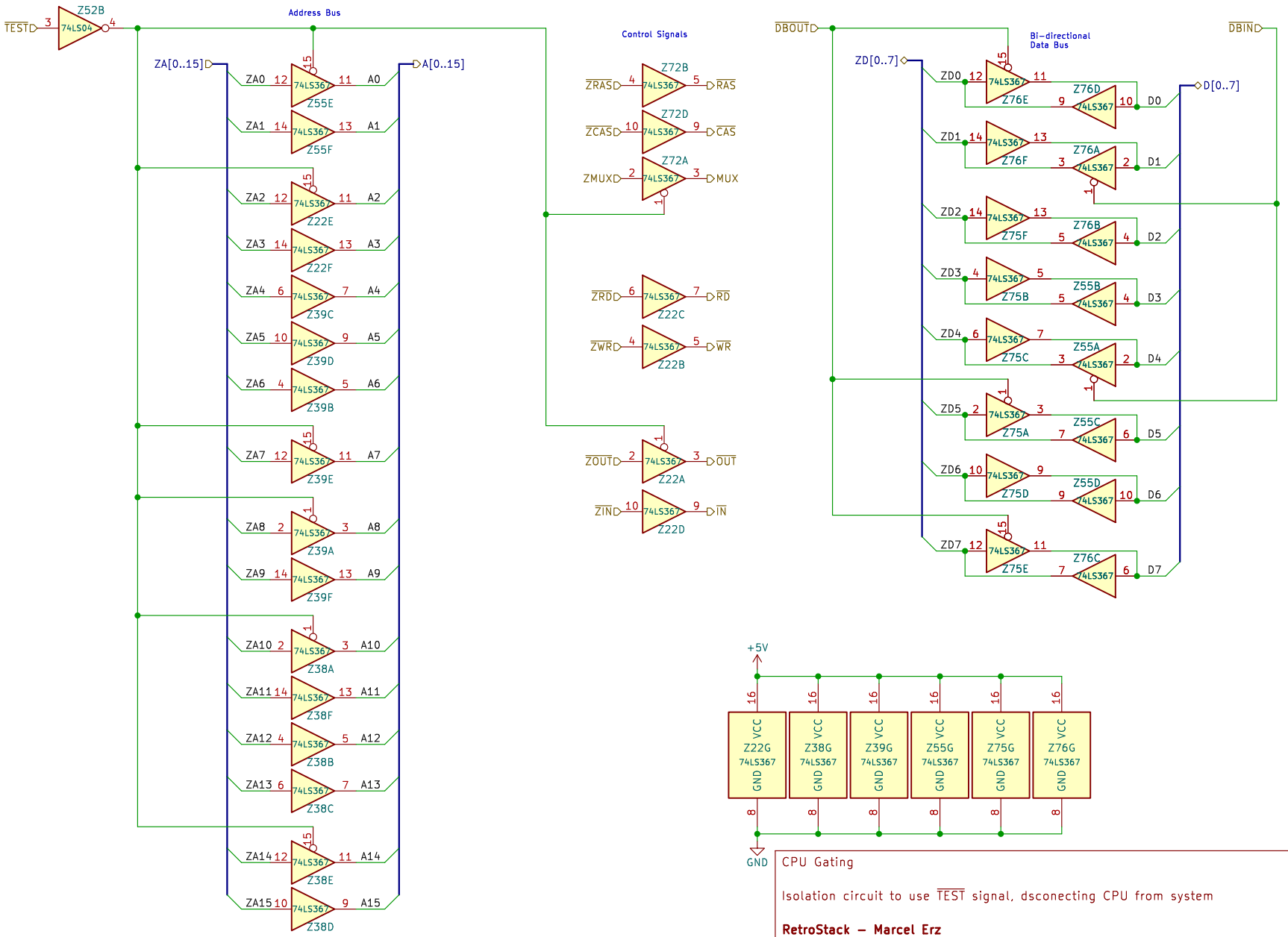
RetroStack – Marcel Erz

Sheet: /CPU/

File: CPU.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 3/21



CPU Gating

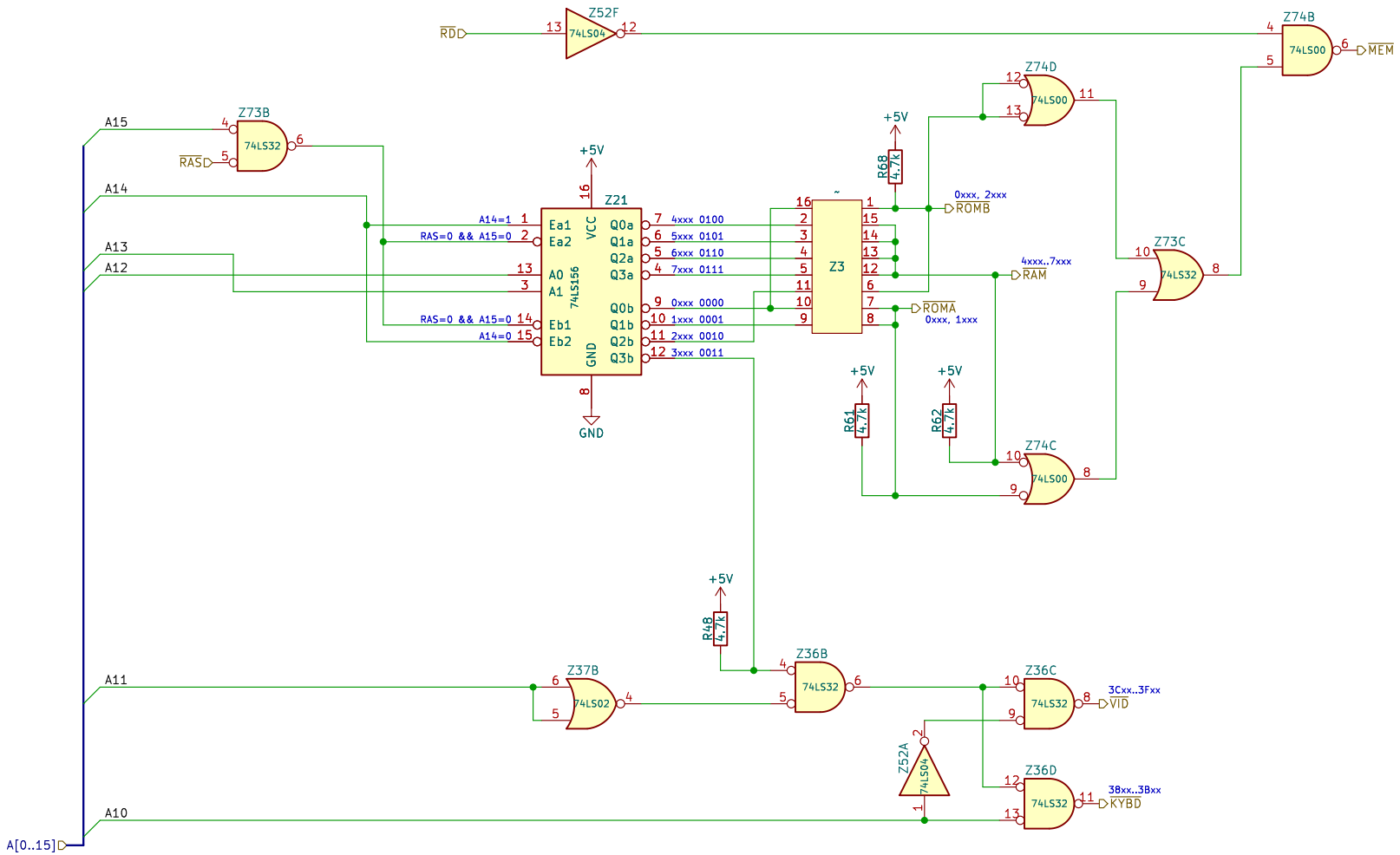
Isolation circuit to use $\overline{\text{TEST}}$ signal, dsconecting CPU from system

RetroStack – Marcel Erz

Sheet: /CPU Gating/
File: CPUGating.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 4/21



Address Decoder

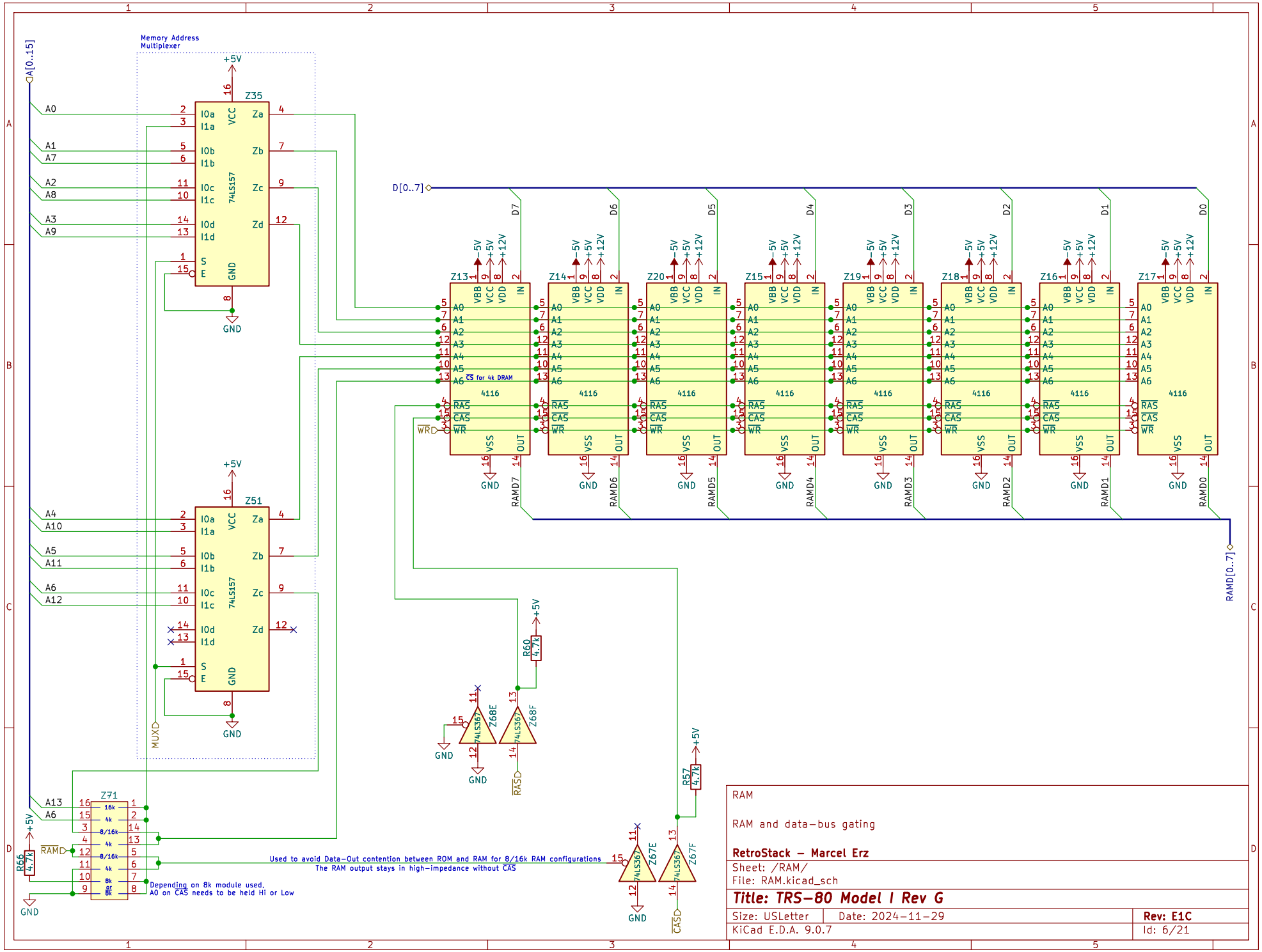
Address decoder circuit for main board components (except IO)

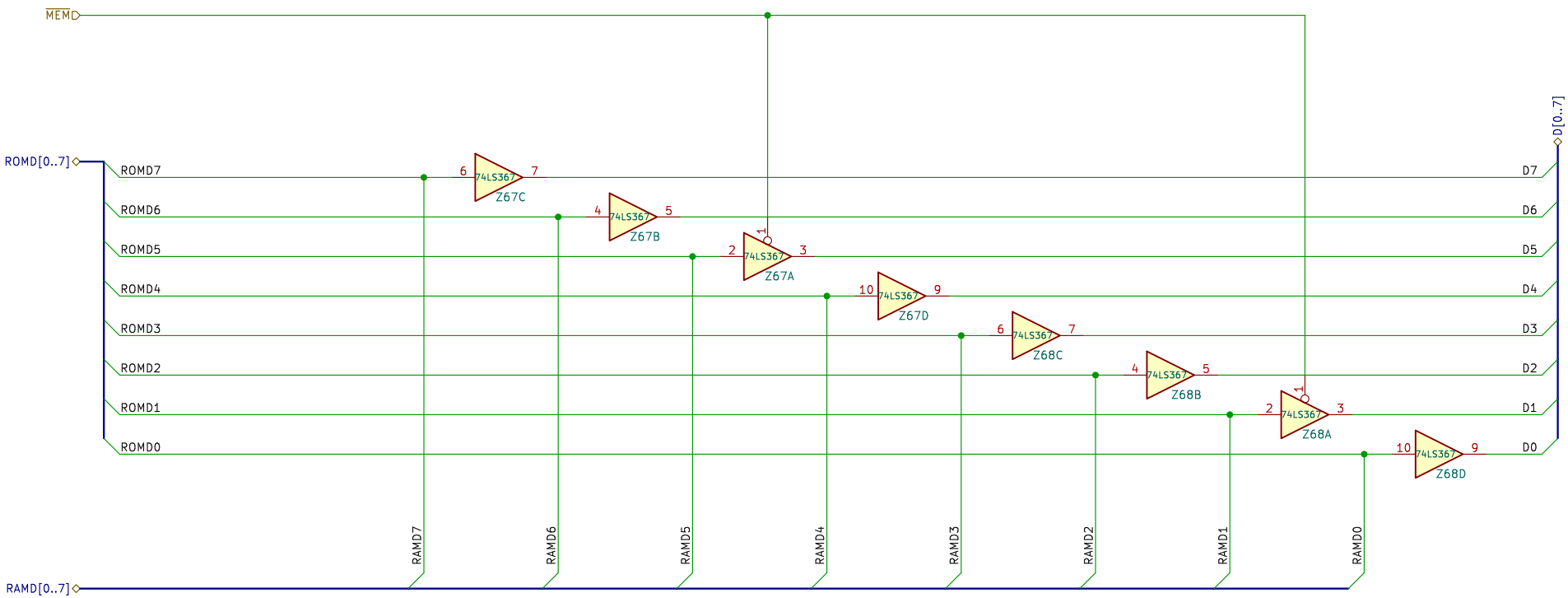
RetroStack – Marcel Erz

Sheet: /Address Decoder/
File: AddressDecoder.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 5/21


Memory Address Multiplexer
A[0..15]
Z35
74LS157
Z51
74LS157
MUXD
D[0..7]
Z13
Z14
Z20
Z15
Z19
Z18
Z16
Z17
4116
CS for 4k DRAM
WR
RAMD7
RAMD6
RAMD5
RAMD4
RAMD3
RAMD2
RAMD1
RAMD0
RAMD[0..7]
R60 4.7k
R57 4.7k
R66 4.7k
Z68E
Z68F
Z67E
Z67F
74LS367
RAS
CAS
Z71
RAM
Used to avoid Data-Out contention between ROM and RAM for 8/16k RAM configurations
The RAM output stays in high-impedance without CAS
Depending on 8k module used,
A0 on CAS needs to be held Hi or Low
RAM
RAM and data-bus gating
RetroStack – Marcel Erz
Sheet: /RAM/
File: RAM.kicad_sch
Title: TRS-80 Model I Rev G
Size: USLetter
Date: 2024-11-29
Rev: E1C
KiCad E.D.A. 9.0.7
Id: 6/21



RAM–ROM Interface

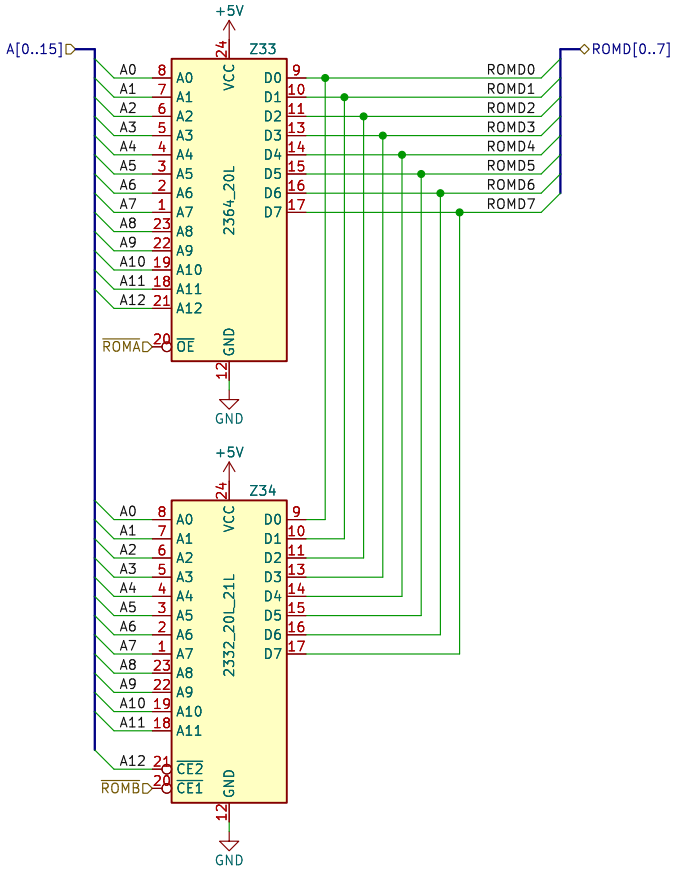
Connects RAM & ROM to the data bus

RetroStack – Marcel Erz

Sheet: /RAM–ROM Interface/
File: RAM_ROM_Interface.kicad_sch

Title: TRS–80 Model I Rev G

Size: USLetter	Date: 2024–11–29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 7/21



Level I ROMs:
- Intel 2616 EPROM (White Crystal Window)
- National MM2316 ROM (Colored Ceramic) (R/D=ROM A, S/D=ROM B)
- Motorola 7800 Series ROMs (7807=ROM A, 7804=ROM B)
- Motorola 7800 Series ROMs (7809=ROM A & B)

Level II ROMs:
- Generic 3 ROMs on separate board with logic chip (each 32kbit/4kb)
- NEC 8043364 & 8043732 (64kbit/8kb & 32kbit/4kb) (early 1979)
- NEC 8044364 & 8044732 (64kbit/8kb & 32kbit/4kb) (early 1980)

ROM

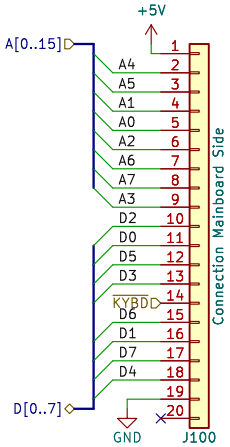
ROM circuit

RetroStack – Marcel Erz

Sheet: /ROM/
File: ROM.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 8/21



Keyboard & Interface

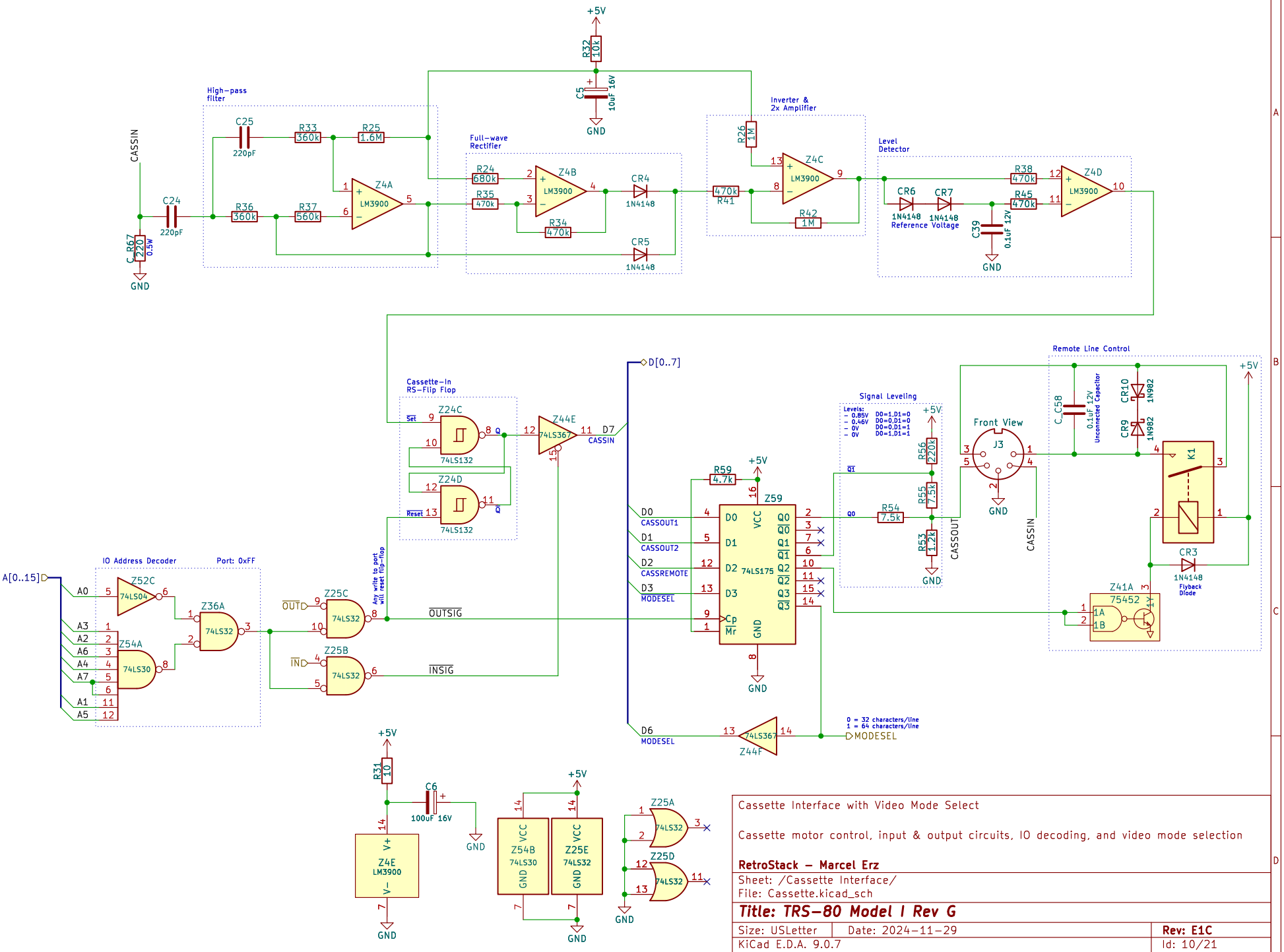
Keyboard interface circuit

RetroStack – Marcel Erz

Sheet: /Keyboard/
File: Keyboard.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 9/21



Cassette Interface with Video Mode Select

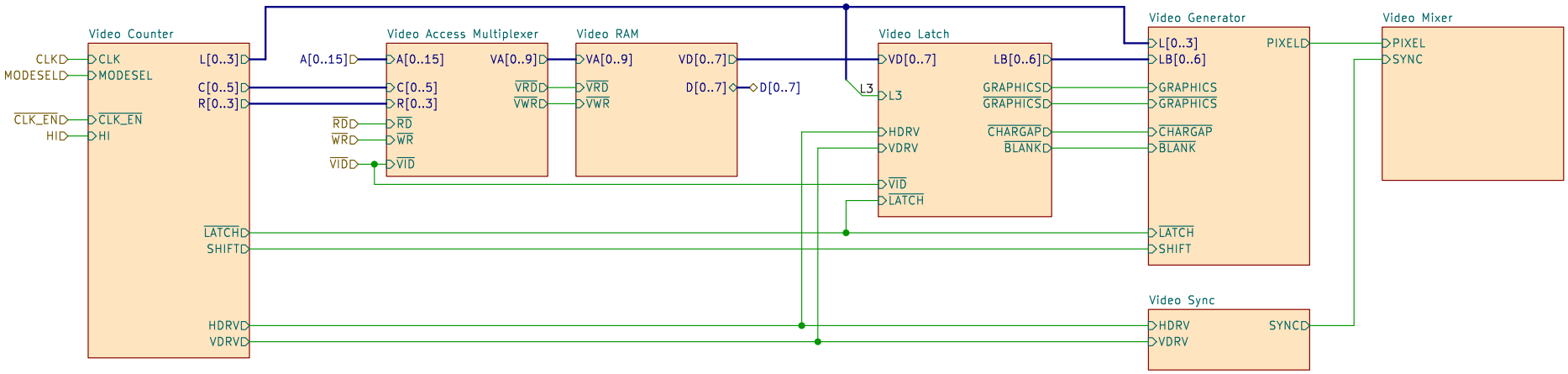
Cassette motor control, input & output circuits, IO decoding, and video mode selection

RetroStack – Marcel Erz

Sheet: /Cassette Interface/
File: Cassette.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 10/21



Video Overview

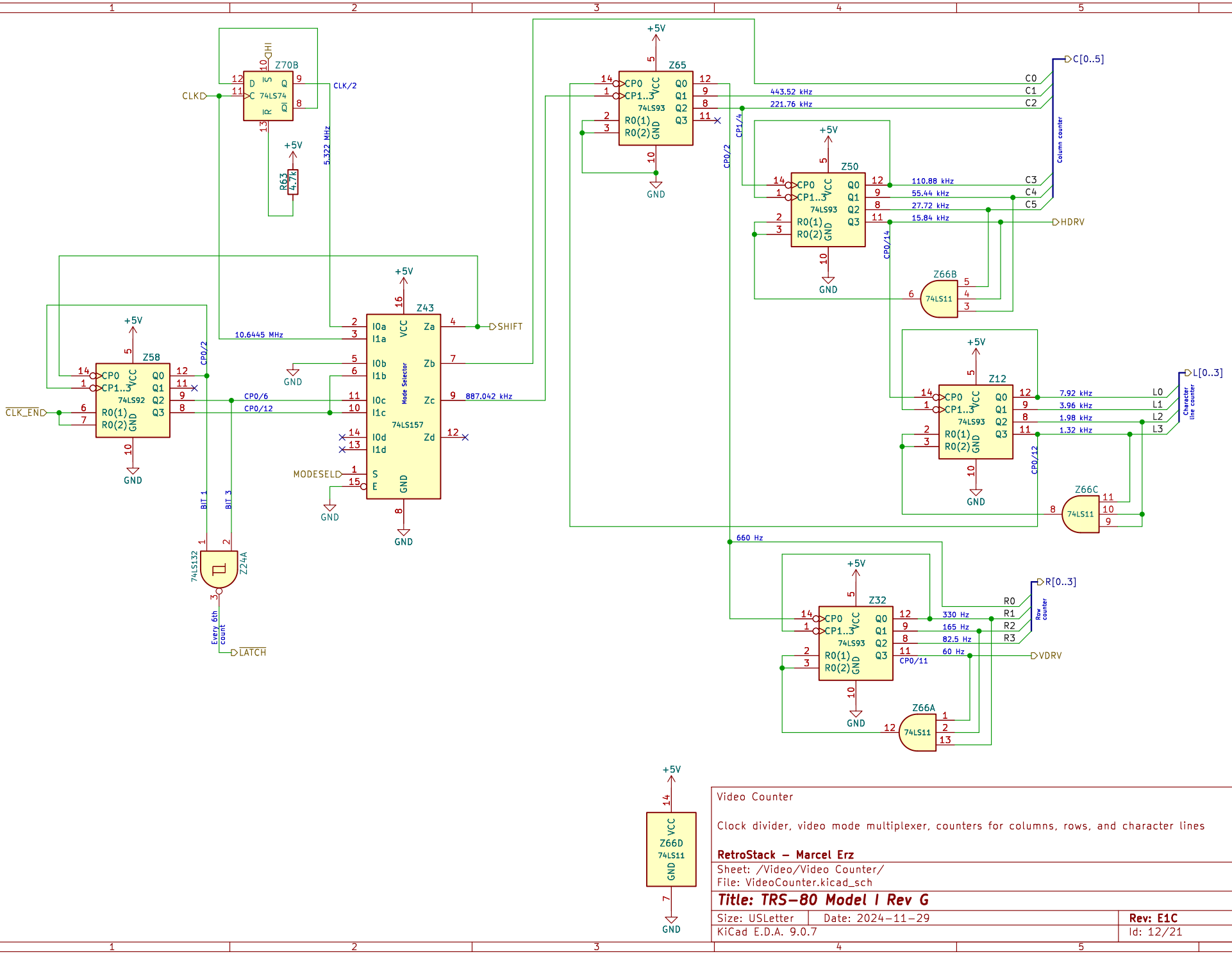
All video components of system

RetroStack – Marcel Erz

Sheet: /Video/
File: Video.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 11/21



Video Counter

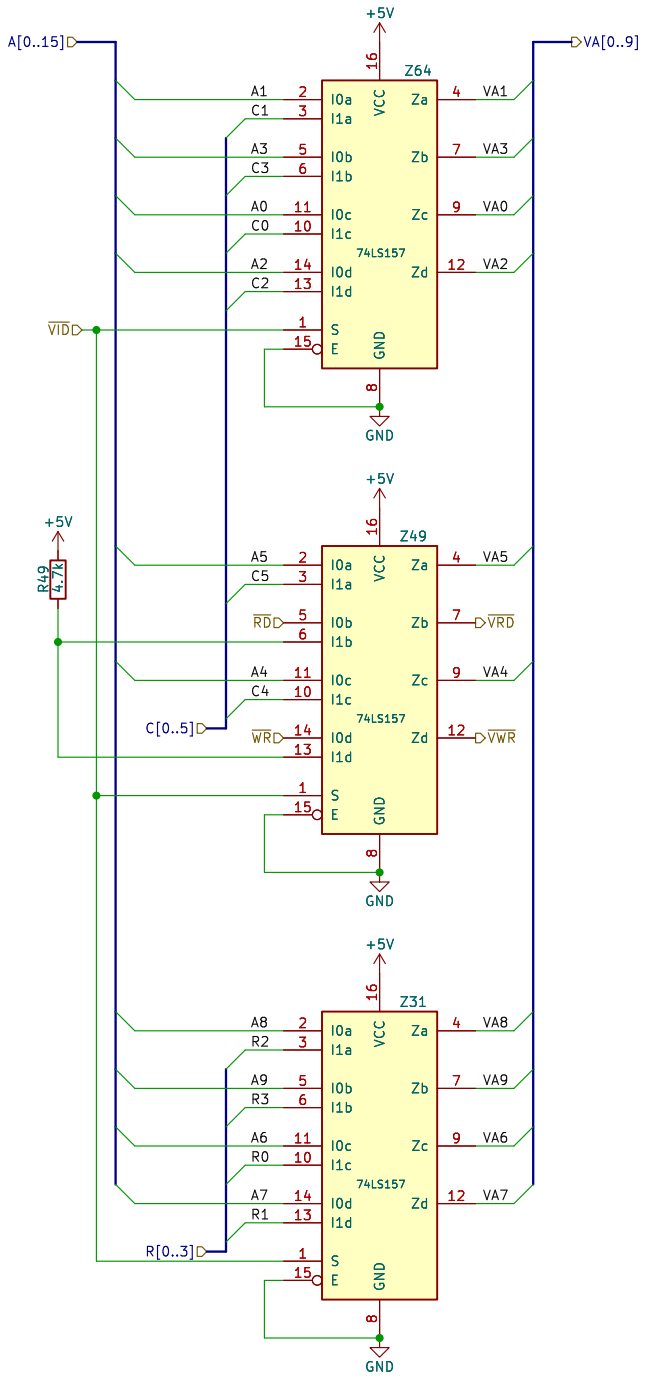
Clock divider, video mode multiplexer, counters for columns, rows, and character lines

RetroStack – Marcel Erz

Sheet: /Video/Video Counter/
File: VideoCounter.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 12/21



Video Access Multiplexer

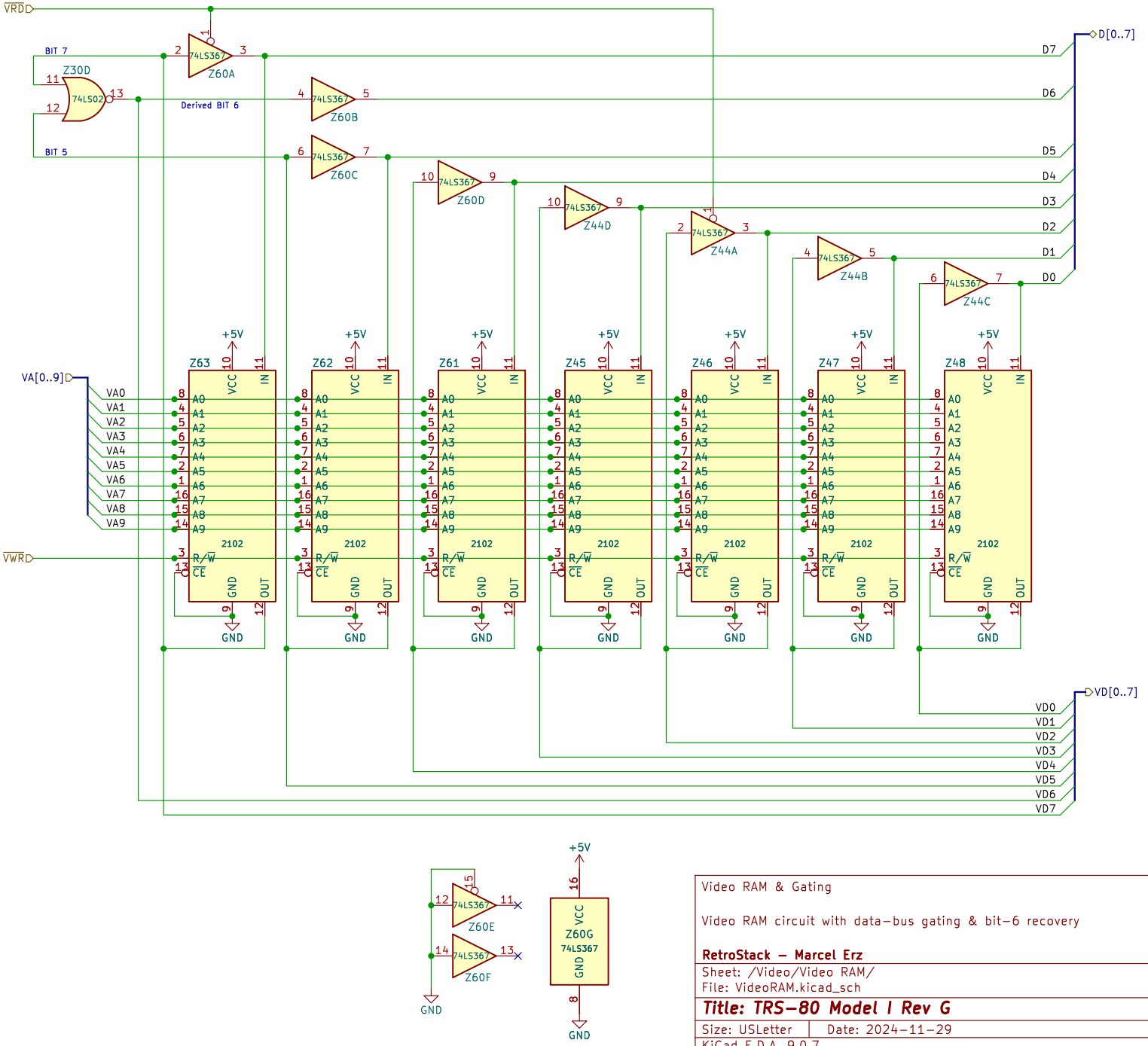
Multiplexer for video RAM access of CPU or video circuit

RetroStack – Marcel Erz

Sheet: /Video/Video Access Multiplexer/
File: VideoAccessMultiplexer.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 13/21



Video RAM & Gating

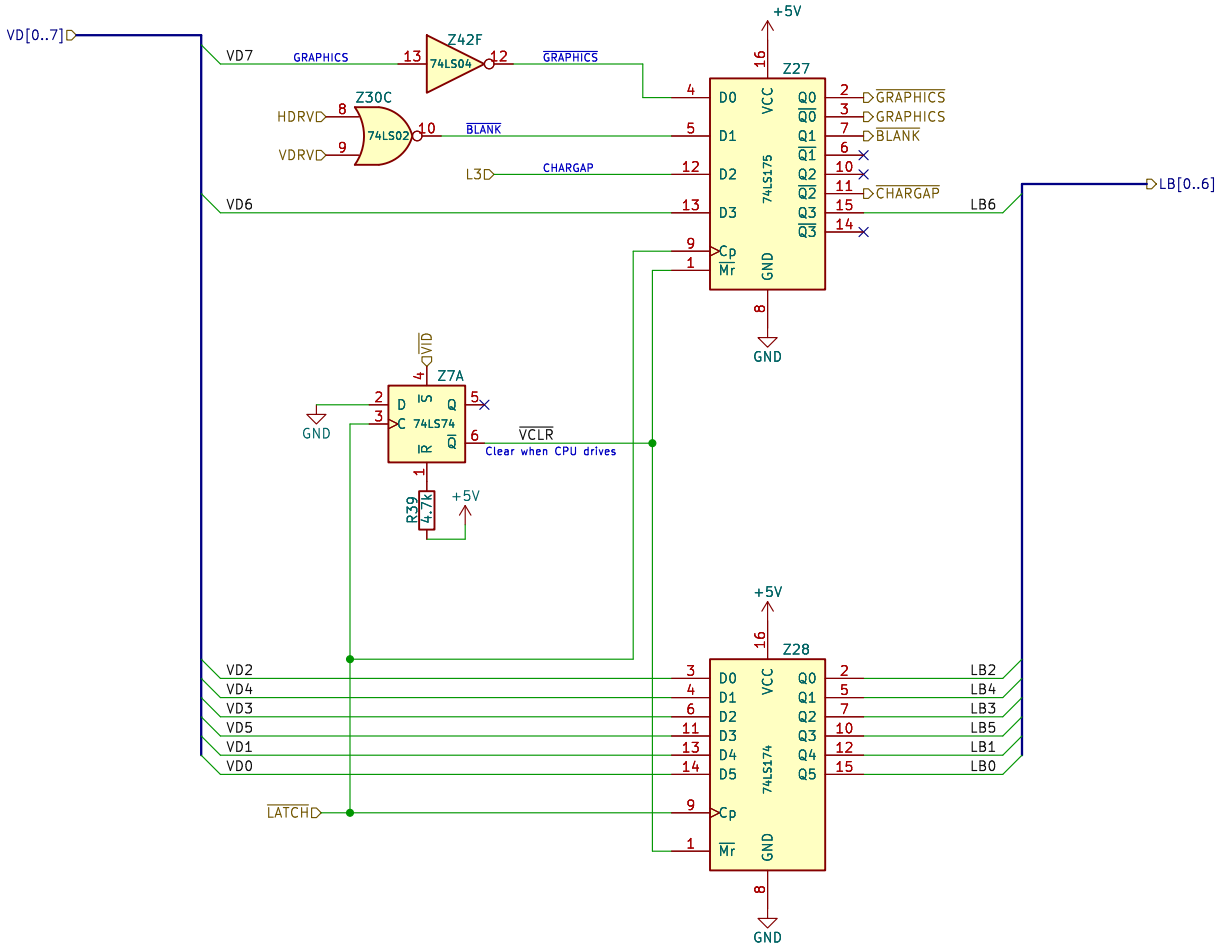
Video RAM circuit with data-bus gating & bit-6 recovery

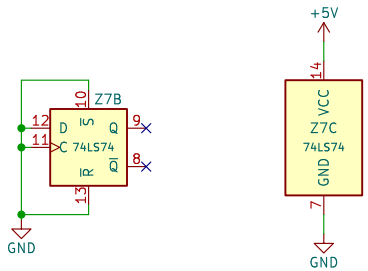
RetroStack – Marcel Erz

Sheet: /Video/Video RAM/
File: VideoRAM.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 14/21





Video Latch

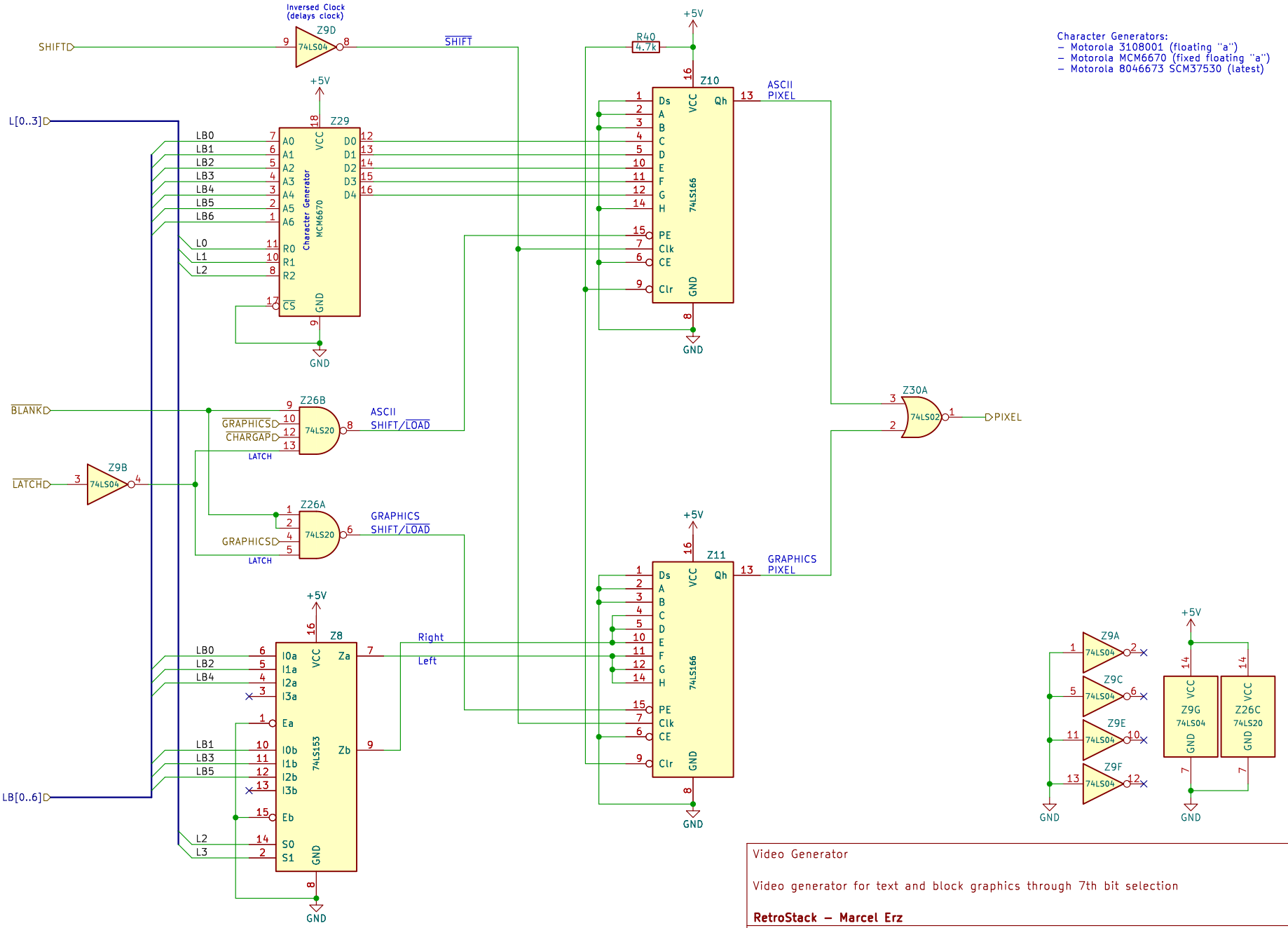
Circuit to latch video content addressed in video RAM

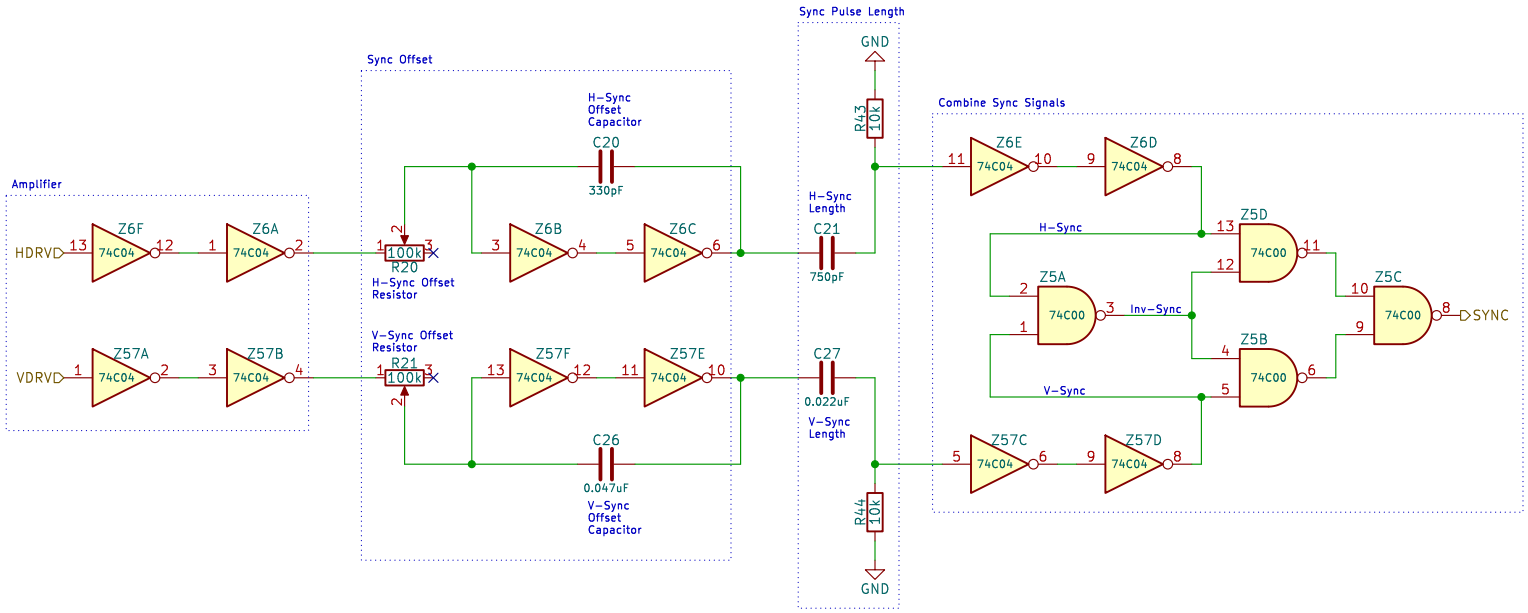
RetroStack – Marcel Erz

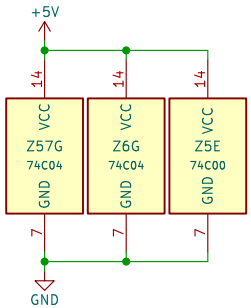
Sheet: /Video/Video Latch/
File: Video Latch.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 15/21


Inversed Clock
(delays clock)
Z9D
74LS04
SHIFT
SHIFT
+5V
R40
4.7k
Z10
74LS166
ASCII
PIXEL
Z29
Character Generator
MCM6670
L[0..3]
LB0
LB1
LB2
LB3
LB4
LB5
LB6
L0
L1
L2
GND
BLANK
Z26B
74LS20
GRAPHICS
CHARGAP
LATCH
ASCII
SHIFT/LOAD
Z9B
LATCH
Z26A
GRAPHICS
SHIFT/LOAD
Z30A
74LS02
PIXEL
Z11
74LS166
GRAPHICS
PIXEL
Z8
74LS153
Right
Left
LB[0..6]
L2
L3
Z9A
Z9C
Z9E
Z9F
Z9G
Z26C
VCC
GND
Character Generators:
– Motorola 3108001 (floating "a")
– Motorola MCM6670 (fixed floating "a")
– Motorola 8046673 SCM37530 (latest)
Video Generator
Video generator for text and block graphics through 7th bit selection
RetroStack – Marcel Erz
Sheet: /Video/Video Generator/
File: VideoGen.kicad_sch
Title: TRS-80 Model I Rev G
Size: USLetter
Date: 2024-11-29
Rev: E1C
KiCad E.D.A. 9.0.7
Id: 16/21





Video Sync

Circuit to combine sync signals and defining front- and back-porches in video signal

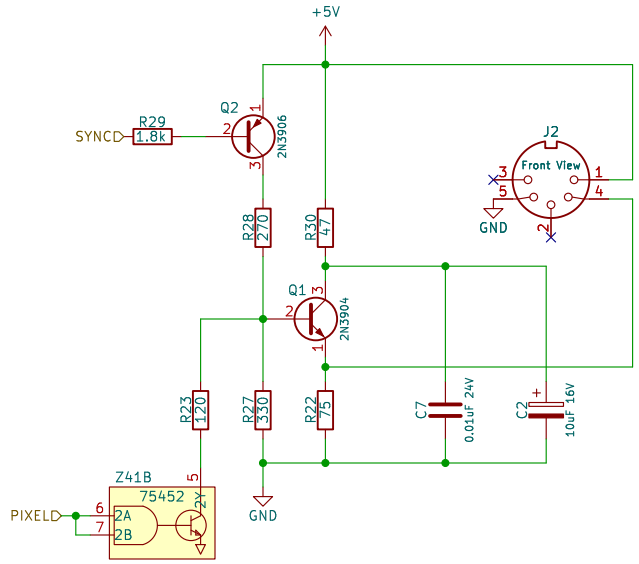
RetroStack - Marcel Erz

Sheet: /Video/Video Sync/

File: VideoSync.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 17/21



Q2 open, Z41 any => 0V (sync level)
Q2 closed, Z41 closed => 1.23V (black level) – R28 with series resistors R23 & R27
Q2 closed, Z41 open => 2.75V (white level) – Only R27 & R28

Video Mixer

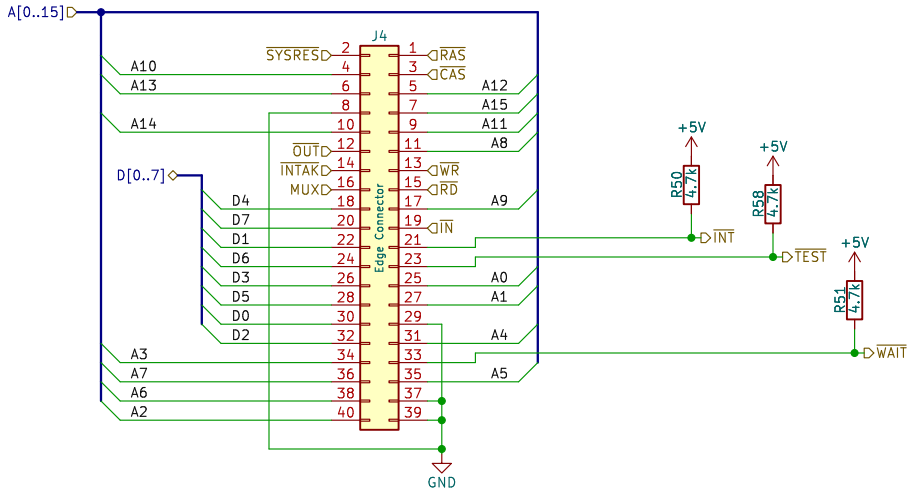
Mixing circuit for sync and pixel data

RetroStack – Marcel Erz

Sheet: /Video/Video Mixer/
File: VideoMixer.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 18/21



Card-Edge Interface

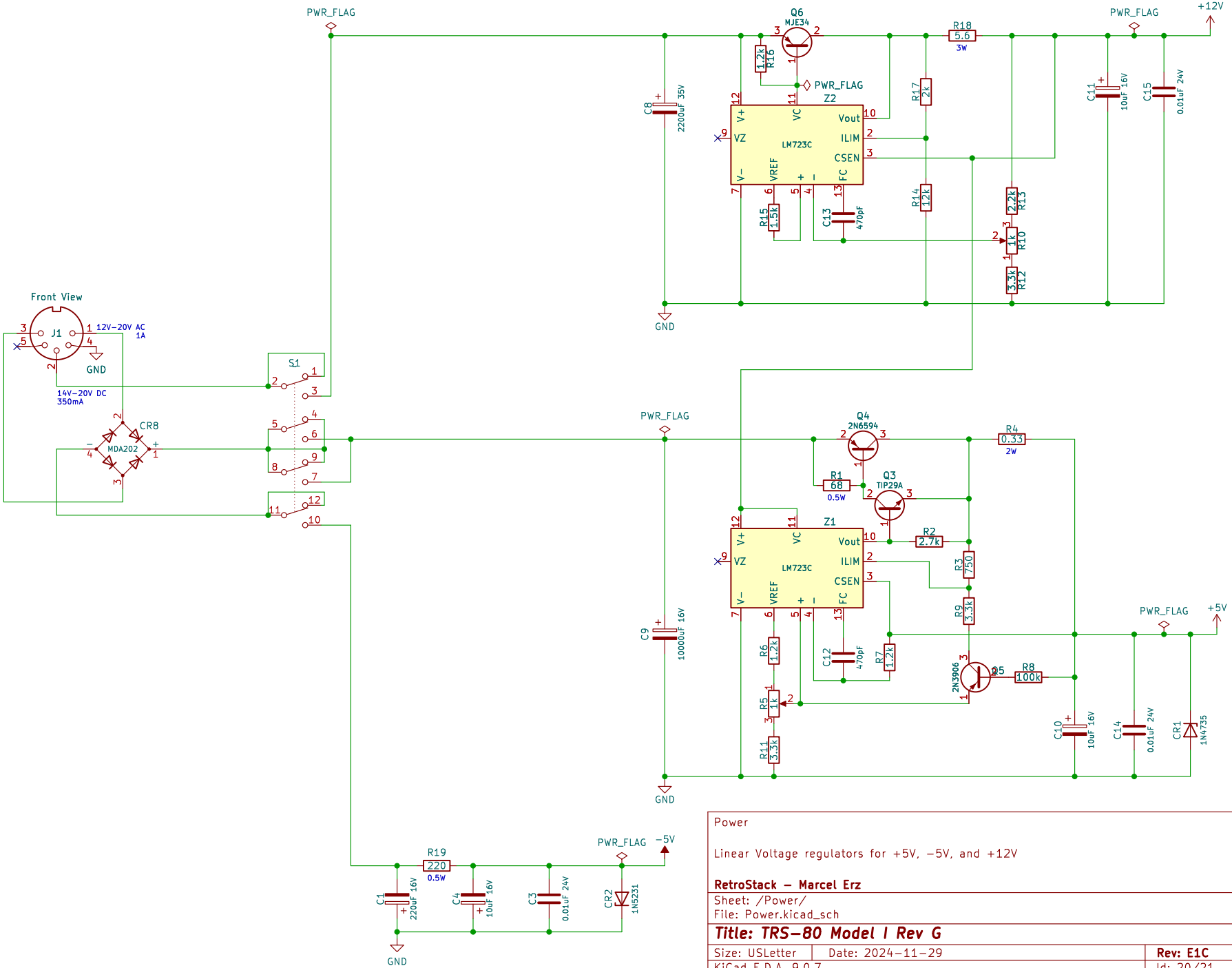
Unbuffered system interface with pull-up circuity for inputs

RetroStack – Marcel Erz

Sheet: /Card-Edge Interface/
File: CardEdgeInterface.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 19/21



Power

Linear Voltage regulators for +5V, −5V, and +12V

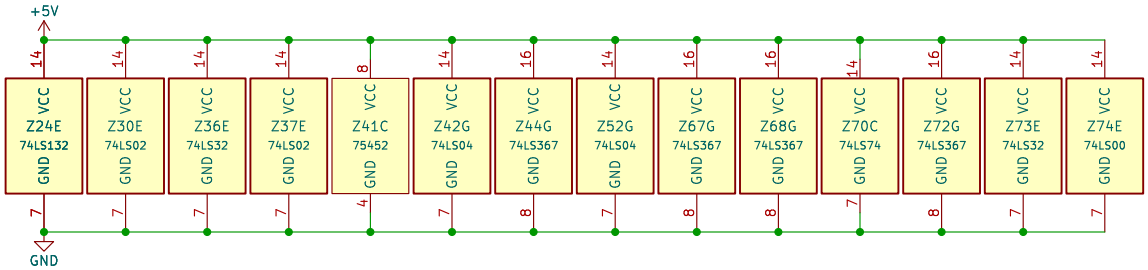
RetroStack – Marcel Erz

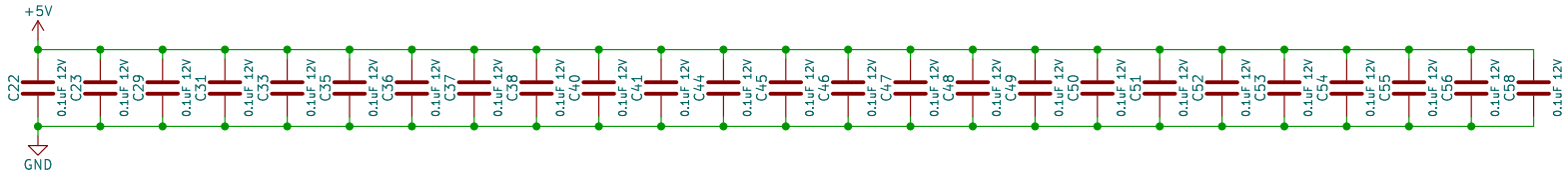
Sheet: /Power/

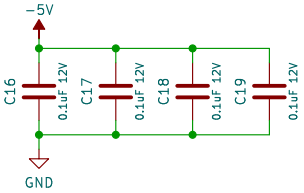
File: Power.kicad_sch

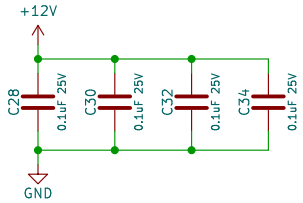
Title: TRS-80 Model I Rev G

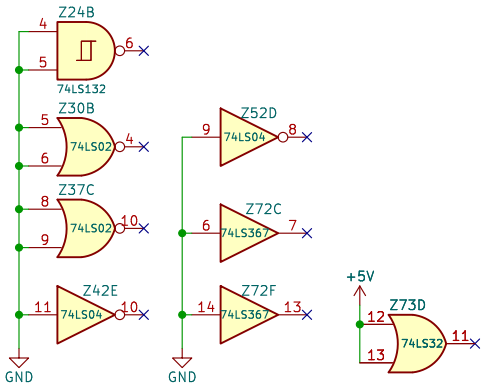
Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 20/21











Capacitors & Others

Main Units, Capacitors, and unused units

RetroStack – Marcel Erz

Sheet: /Capacitors/

File: Capacitors.kicad_sch

Title: TRS-80 Model I Rev G

Size: USLetter	Date: 2024-11-29	**Rev: E1C**
KiCad E.D.A. 9.0.7		Id: 21/21